

DOM207-DOM3007-DOM6401: MP1

BACKGROUND

In this project I was a senior analytics consultant in 2008 for an elite restaurant chain that operated primarily at major entertainment places in the US, such as Las Vegas and the Atlantic City, among others. It was a multinational firm which also had branches and restaurants in several cities in the Western Europe.

The project that you'll be doing here is a much simplified and scaled-down version of the actual project which was very big. The original data file was huge, but I've given you a much, much shortened version of it containing only a few fields from the original file. However, the data is real.

DIRECTIONS:

- ✓ In this project, assume you're members of the DSA team hired in a contract by a restaurant business named Foodie India (that's your client)
- ✓ The objective of this project is to recommend to Foodie's management some **prospective and viable business propositions for expansion** based only on the data that have been provided to you
- ✓ The data set *Restaurant.xlsx* belongs to Foodie. It's their proprietary data, so **you're NOT permitted to add any new data to it**
- ✓ You must bring to bear all the knowledge to deliver viable propositions for immediate business expansion for Foodie

WORK AND SUBMISSION REQUIREMENTS AND GUIDELINES:

- ✓ For this project, you must work in a group of **2 (ideally, for best learning and work experience) but no more than 3 students per group**.
- ✓ Each group must submit only one project report – one member from each group submit the report and all attendant files to BB.
- ✓ Each submission must contain the following items:
 - A report in word doc or PDF with an appropriate **title page**. This page should contain a title for the project, **your name and the name(s) of your team members**, your roll #, course name and **code**
 - **Like a corporate project report, your report must be divided into the following sections:**
 - **Section:** Introduction (say what you're set out to achieve in this project)
 - **Section:** Data (pre)processing (provide details of all data cleansing steps you've undertaken)
 - **Section:** Data visualization (provide some worthwhile charts, diagrams, etc. for visualization – do **not** show useless diagrams – you'll be penalized if you do)
 - **Section:** Analysis (provide **complete** statistical analysis with justifications for all of your findings). **You MUST perform some HT.**
 - You may also want to insert some helpful screenshots of your work (C/P from your output). These must be relevant and convincing
 - **Section: Discussions and Recommendations.** **Your final recommendations with proper justifications** (based on your findings, **not speculations**)
 - **Section: References**

- Submit the project report (Word doc or PDF) along with your R or Python files containing the complete analysis (**project files and .R or .py files**)
- There's no fixed requirement for word or page limit for the report. However, the material in it should be substantial and meaningful, exactly like a project report that you're required to submit to a company's upper management

NOTE: Submission is completely your responsibility.

VERY IMPORTANT TO REMEMBER

- If you fail to adhere to the above directions, **you'll be penalized**. The project should be as formal as possible
- **Project submission deadline is inflexible and strictly enforced!!!**
- **All submissions are on BB. TA will NOT accept any email submissions. Submit well ahead of the due date/time to avoid last-minute glitches. Excuses for late submission on BB even by a minute are considered unprofessional – they're UNACCEPTABLE by our standards**
- **We follow corporate practice in all MPs for this course**

Explanation of fields in the data file [Restaurant.xlsx](#)

- *Amount*: Total bill amount for a group of customers (including a single individual customer), called a party, at a single meal
- *Tip*: Amount of tips given by a party to the waiter who served that party
- *Gender*: Gender of the waiter who served a party
- *Smoker*: Whether any individual in a party smoked or not
- *Day*: Day of the meal (Thursday; Friday; Saturday; Sunday)
- *Time*: Time of the day of the meal (Lunch; Dinner)
- *Partysize*: How many people there were at the party having the meal at the restaurant